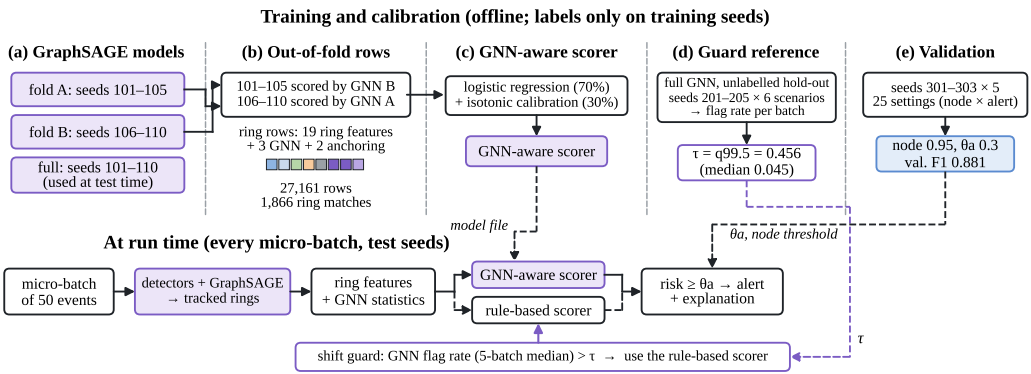
Training and calibration (offline; labels only on training seeds)
(a) GraphSAGE models
fold A: seeds 101–105
fold B: seeds 106–110
full: seeds 101–110
(used at test time)
(b) Out-of-fold rows
101–105 scored by GNN B
106–110 scored by GNN A
ring rows: 19 ring features
+ 3 GNN + 2 anchoring
27,161 rows
1,866 ring matches
(c) GNN-aware scorer
logistic regression (70%)
+ isotonic calibration (30%)
GNN-aware scorer
(d) Guard reference
full GNN, unlabelled hold-out
seeds 201–205 × 6 scenarios
→ flag rate per batch
τ = q99.5 = 0.456
(median 0.045)
(e) Validation
seeds 301–303 × 5
25 settings (node × alert)
node 0.95, θa 0.3
val. F1 0.881
model file
θa, node threshold
At run time (every micro-batch, test seeds)
micro-batch
of 50 events
detectors + GraphSAGE
→ tracked rings
ring features
+ GNN statistics
GNN-aware scorer
rule-based scorer
risk ≥ θa → alert
+ explanation
shift guard: GNN flag rate (5-batch median) > τ → use the rule-based scorer
τ